



A comparative analysis of TE and TM guided modes in Ti-diffused Er:LiNbO₃ waveguide lasers

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Abstract. A detailed comparative analysis of TE and TM guided-modes in Z-cut Ti-diffused Er:LiNbO₃ strip waveguide lasers has been performed at laser emission wavelengths 1532, 1563, 1576 nm and pump wavelengths 1480 and 980 nm by the variational method. First, the mode size and corresponding effective refractive index at these five wavelengths as a function of initial Ti-strip width W , diffusion temperatures T , initial Ti-strip thickness H and diffusion time t were calculated for several lower-order TE and TM modes; their characteristics and the single mode condition were discussed comparatively. Then, the plots of effective pump area and coupling efficiency between the laser and the pump modes under different pump and output mode polarizations were analysed. Finally, the coupling loss of a joint between the waveguide and a fibre in terms of these diffusion parameters were also calculated at these wavelengths. Numerical results are in good agreement with previously published experimental data.

1. Introduction

There has been an increasing interest in rare-earth doped, optically pumped waveguide amplifiers and lasers with the dramatic increase of functionality for waveguide LiNbO₃ optical devices. Recent developments based on this material have demonstrated the potential for commercially attractive lasers and nonlinear optical devices which are compact, efficient, inexpensive and manufacturable. Some devices have shown potential commercial prospects, e.g. (1) diffusion-doped Er:LiNbO₃ waveguide laser sources that are packaged, pigtailed and compatible with fibre systems and (2) bulk-doped Nd: and Er:LiNbO₃ waveguide lasers illustrating active mode-locking and Q-switching.

Erbium doping is of great interest as it leads to optical amplification and lasing in the wavelength range 1530-1676 nm, which is most important for optical fibre communication. Recently, continuous wave (CW) emitting, free running Ti-diffused Er:LiNbO₃ waveguide lasers operating at wavelengths 1532, 1563 and 1576 nm [1-5] and integrated devices such as mode-locked [6, 7], tunable [8] and DBR waveguide lasers [9] have been successfully demonstrated. Quantitative models were also proposed to improve the understanding of the physical phenomena involved [3], [10-12].

Mode size is an important parameter. It is closely related to many physical quantities of the waveguide laser such as effective pump area (or effective mode volume), which is directly proportional to the threshold pump power, coupling efficiency between laser and pump modes, which is proportional to slope efficiency, and coupling loss between a waveguide and a fibre. It can be also used to determine the effective refractive index of a guided mode and the single mode propagation condition of a waveguide with the variational technique.

In this paper, a detailed comparative analysis of transverse electric (TE) and transverse magnetic (TM) guided modes in Ti-diffused Er:LiNbO₃ waveguide lasers has been carried out at emission wavelengths 1532, 1563 and 1576 nm and pump wavelengths 1480 and 980 nm using the variational method. Er:LiNbO₃ waveguide lasers pumped at 980 nm suffer from excited state absorption (ESA) and serious photorefractive damage, therefore, pumping at 1480 nm is preferred. Huang and McCaughan [13] have reported a 980 nm pumped Er:LiNbO₃ amplifier in which the detrimental effect of photorefractive damage on the amplifier gain is evident. However, since the photorefractive effect could be reduced by co-doping with the appropriate quantity of MgO, and if the excited state absorption could be tolerated, then pumping at 980 nm is also possible. Another way of considerably reducing this optical damage is by propagating along the *Z* axis; consequently, stable CW lasers in X-cut, *Z*-propagating Ti-diffused Nd: and Er:LiNbO₃ waveguides with greatly reduced photorefractive instability have been successfully demonstrated by Amin *et al.* [5]. For completeness and comparison with 1480 nm, pumping at 980 nm is also considered here. In previous papers [14, 15], the preliminary theoretical results of TM guided modes at 1530 nm have been reported. In comparison with this work, the present study has following innovations.

- (1) The other two emission wavelengths 1563 and 1576 nm were considered.
- (2) Both TE and TM guided modes were comparatively analysed in detail.
- (3) In addition to the TM-pumping/TM-output scheme (for convenience, hereafter, we refer to it as TM-TM) considered earlier in [14], the other three possible schemes TM-TE, TE-TM and TE-TE were also compared.
- (4) Another controllable diffusion parameter, i.e. diffusion time *t*, was considered.
- (5) Dependence of coupling loss of a joint between the waveguide and a fibre on initial strip width *W*, diffusion temperature *T*, initial strip thickness *H* and diffusion time *t* were also calculated at the above five wavelengths.
- (6) The single mode conditions of both TE and TM modes were collected for all of the wavelengths considered.

2. Numerical results and discussions

2.1. Mode size

For arbitrary order modes (TE_{*n*0} and TM_{*n*0}) guided in a *Z*-cut Ti:LiNbO₃ waveguide, considering the mode field distribution and the index profile of the waveguide as employed in [15], the variational equations for the propagation constant can be expressed as:

for TE_{*n*0} modes,

$$\beta^2 = k_0^2 n_{\text{ob}}^2 + \frac{2(2^{1/2})(n_{\text{os}}^2 - n_{\text{ob}}^2)k_0^2}{W_x} \left(1 + \frac{W_y^2}{2d_{y_0}^2}\right)^{-3/2} Q - \frac{2n+1}{W_x^2} - \frac{3}{W_y^2}; \quad (1)$$

for TM_{n0} modes,

$$\beta^2 = k_0^2 n_{\text{eb}}^2 + \frac{2(2^{1/2})(n_{\text{es}}^2 - n_{\text{eb}}^2)/k_0^2}{W_x} \left(1 + \frac{W_y^2}{2d_{y_e}^2}\right)^{-3/2} Q - \frac{2n+1}{W_x^2} - \left(\frac{n_e}{n_o}\right)^2 \frac{3}{W_y^2}; \quad (2)$$

where

$$Q = \int_0^{x_z} g\left(\frac{2x}{W}\right) \psi_n^2\left(\frac{2^{1/2}x}{W_x}\right) dx, \quad (3)$$

in which the upper limit of the integral x_z satisfies $g(2x/W) = 0$. x and y are the coordinates along the width and depth directions of the waveguide, respectively. $n_{\text{eb,ob}}$ and $n_{\text{es,os}}$ are substrate and surface indexes, respectively, W_x and W_y are the so-called mode sizes. Their optimum values correspond to the largest value of β^2 , namely β_m^2 , then the effective refractive index can be determined by the following expression:

$$N_{\text{eff}} = \frac{\lambda}{2\pi} \beta_m. \quad (4)$$

We restrict our calculations to a Z-cut Ti-diffused LiNbO₃ waveguide and the calculations were performed for TE_{n0} and TM_{n0} ($n=0,1,2$) modes at the emission wavelengths 1532, 1563, 1576 nm and pump wavelengths 1480 and 980 nm. The waveguide fabrication parameters employed were taken from [2], i.e. the waveguide (with the length of 52 mm) fabricated by indiffusing 8 μm wide, 95 nm thick Ti-strip at 1030°C for 9 h, instead of the figures 7 μm , 100 nm, 1050°C and 8.3333 h employed in our earlier work [14, 15]. Calculated results have indicated that the mode size increases slowly and almost linearly as the wavelength is increased in the range 1480-1650 nm, therefore, in figures 1(a)-(d) we only show, as a representative, the variations of TE and TM mode size with the strip width W (the other diffusion parameters are unchanged, i.e. $H = 95$ nm, $T = 130^\circ\text{C}$ and $t = 9$ h, and similarly in the following three cases), diffusion temperature T , initial strip thickness H and diffusion time t for the wavelength 1563 nm. Figure 2 is the case for 980 nm. It is evident from these plots that the TE mode sizes reveal similar varying features as those of TM modes. The mode size of the TE mode is obviously larger than that of the TM mode. One can see from equations (1) and (2) that the surface index increment, as the second term in the two equations indicates, is a key factor in determining the mode size. The larger surface index increment should correspond to smaller mode size. As we know, for a Ti-diffused LiNbO₃ waveguide, the ordinary surface index increment is smaller than the extraordinary surface index increment for a given wavelength, since the TE mode is related to the ordinary index n_o , while the TM mode is mainly associated with the extraordinary index n_e , so TE mode sizes are generally larger than those of TM modes. For 1563 nm (see figure 1), both TE and TM mode sizes reveal a stronger dependence on W , T and H , and a relatively weaker dependence on t . As the considered wavelength lowers from 1563 to 980 nm, these dependences become somewhat weaker (see figure 2). Another feature shown by figures 1 and 2 is that for certain diffusion parameters the

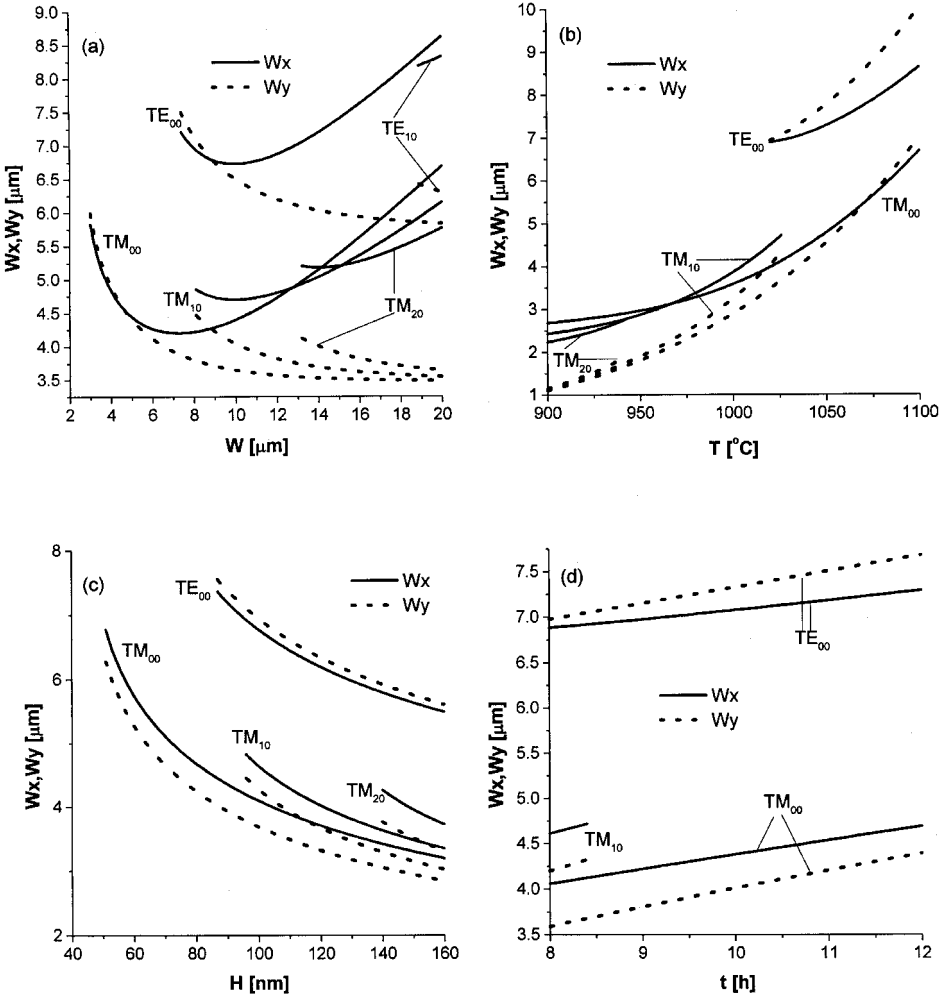


Figure 1. TE and TM mode size at 1563 nm versus (a) Ti-strip width, (b) diffusion temperature, (c) strip thickness and (d) diffusion time.

waveguide is expected to support more TM modes and fewer TE modes (figure 1(b)), especially when the wavelength is in the range 1480-1650 nm. For 1563 nm, the waveguide essentially guides a single TE mode in the studied ranges as illustrated in figure 1.

2.2. Effective refractive index and single-mode condition

The effective refractive index N_{eff} corresponding to the plots of figures 1 and 2 are illustrated in figure 3 (for 1563 nm) and figure 4 (for 980 nm). For the TM mode, the varying characteristic of the index versus these parameters are similar for the two wavelengths, that is, it increases with a rise in W or H , and decreases with increasing T or t . For the TE mode, except for the plots of the index against W or H , the variations are dramatic. For 1563 nm, the TE mode index increases as T or t is increased. For 980 nm, it increases as T is increased from 900 $^{\circ}\text{C}$, near 950 $^{\circ}\text{C}$ it is expected to pass through a maximum, then it decreases gradually as T is

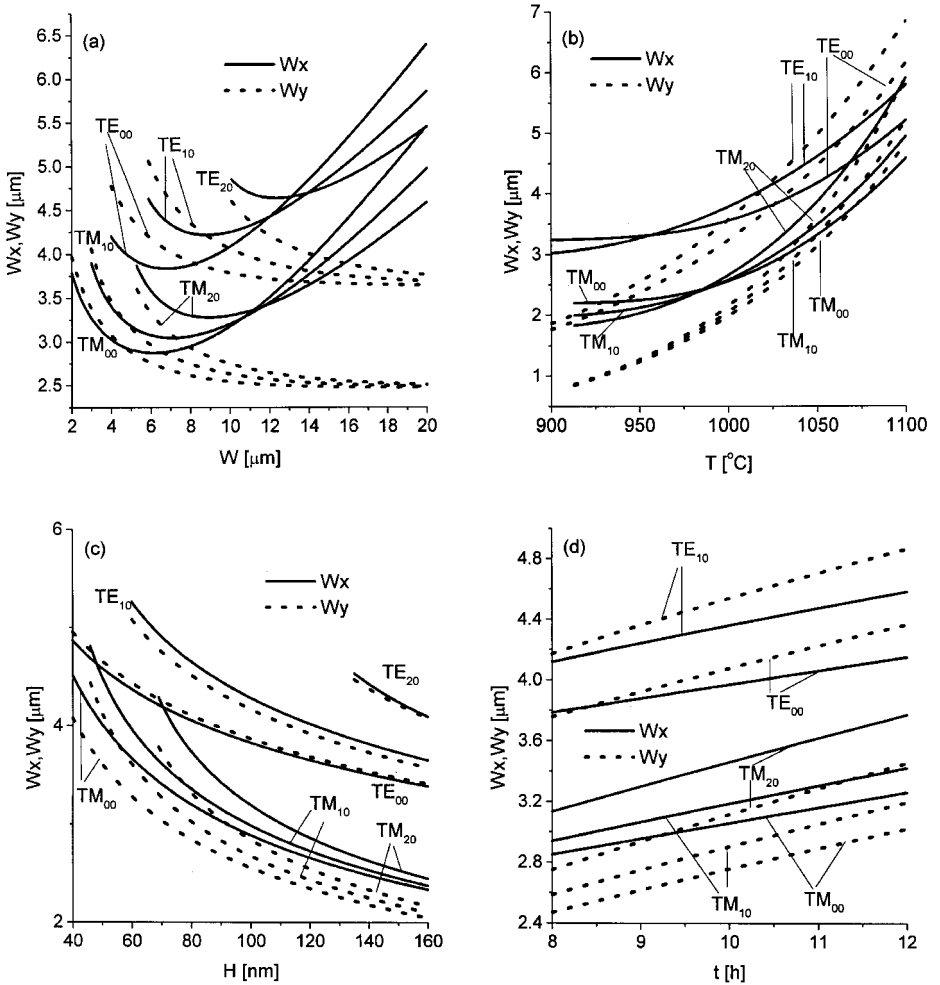


Figure 2. TE and TM mode size at 980 nm versus (a) Ti-strip width, (b) diffusion temperature, (c) strip thickness and (d) diffusion time.

increased further, as shown in figure 4. The plots of index versus t for 980 nm TE modes similarly to the TM plots are almost straight lines, and their varying behaviour is also the same as TM plots.

According to the relation $n_{\text{eb,ob}} < N_{\text{eff}}$, the single-mode condition can be easily determined from the plots of figures 3 and 4. Table 1 summarizes the results for both TE and TM polarization at the emission wavelengths 1532, and 1576 nm, and pumping wavelengths 1480 and 980 nm. One can see from the table that the waveguide fabricated by indiffusing a $8\text{ }\mu\text{m}$ wide, 95 nm thick Ti-strip at 1030°C for 9 h will be approximately single mode for both TE and TM polarization at emission wavelengths 1532, 1563 and 1576 nm, and pumping wavelength 1480 nm (note that at this wavelength the waveguide seems to support TM_{00} and TM_{10} two modes, but the latter is near cut-off). This is consistent with the experimental results reported by Becker *et al.* [2]. For 980 nm, however, this waveguide will

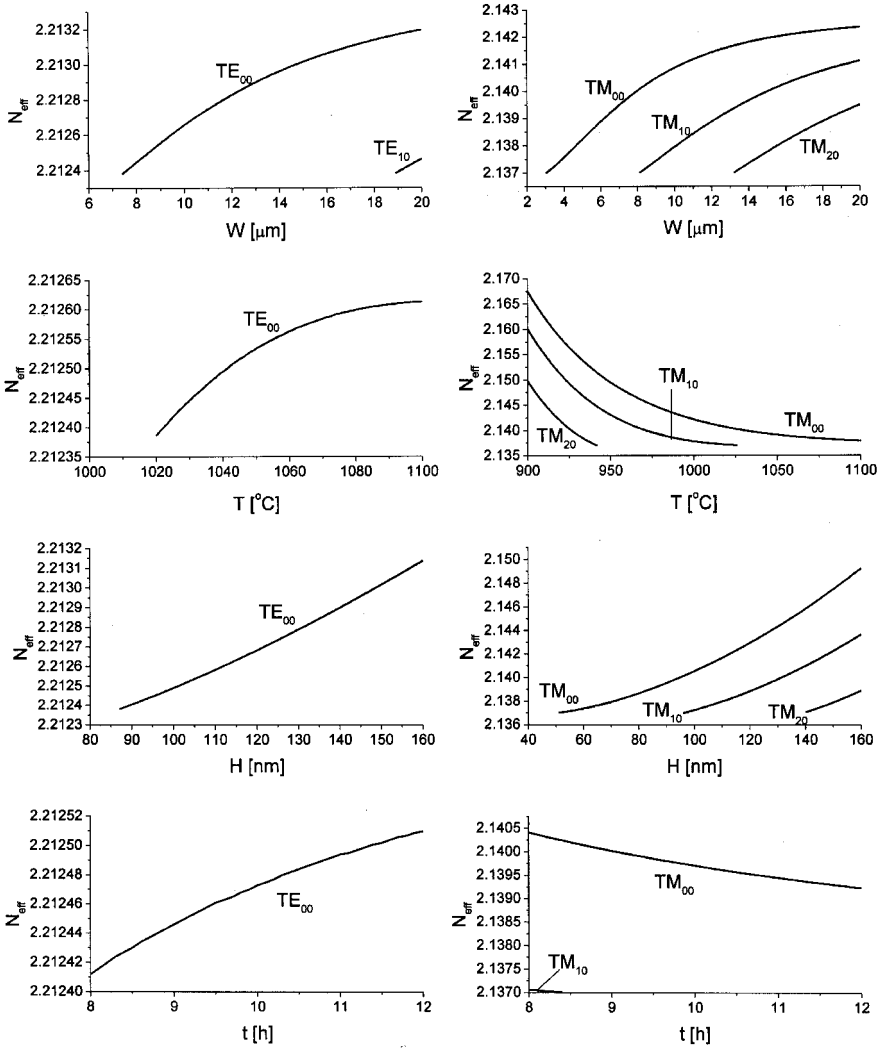


Figure 3. TE and TM mode effective refractive index at 1563 nm versus Ti-strip width, diffusion temperature, strip thickness and diffusion time.

support two TE modes the three TM modes, and this result is qualitatively supported by the experimental results reported by Amin *et al.* [5].

Table 1 also reveals that TE single mode propagation is more easily realized than TM single mode. In fact, for 1480, 1532, 1563 and 1576 nm only TE_{00} modes are supported for the waveguides fabricated by the parameters in all of the ranges considered; comparatively, the TM single mode conditions are more rigorous. As the considered wavelength is lowered to 980 nm, the single mode propagation is difficult for both TE and TM polarization, unless W is less than 6.0 and 3.0 μm , respectively, or H is less than 60 and 46 nm, respectively. Therefore, for 1480 nm pumping, TE and TM fundamental mode laser oscillation and pumping could be readily realized, especially for TE pumping and TE output. For 980 nm pumping, however, in order to guarantee the single-mode laser operation and single-mode

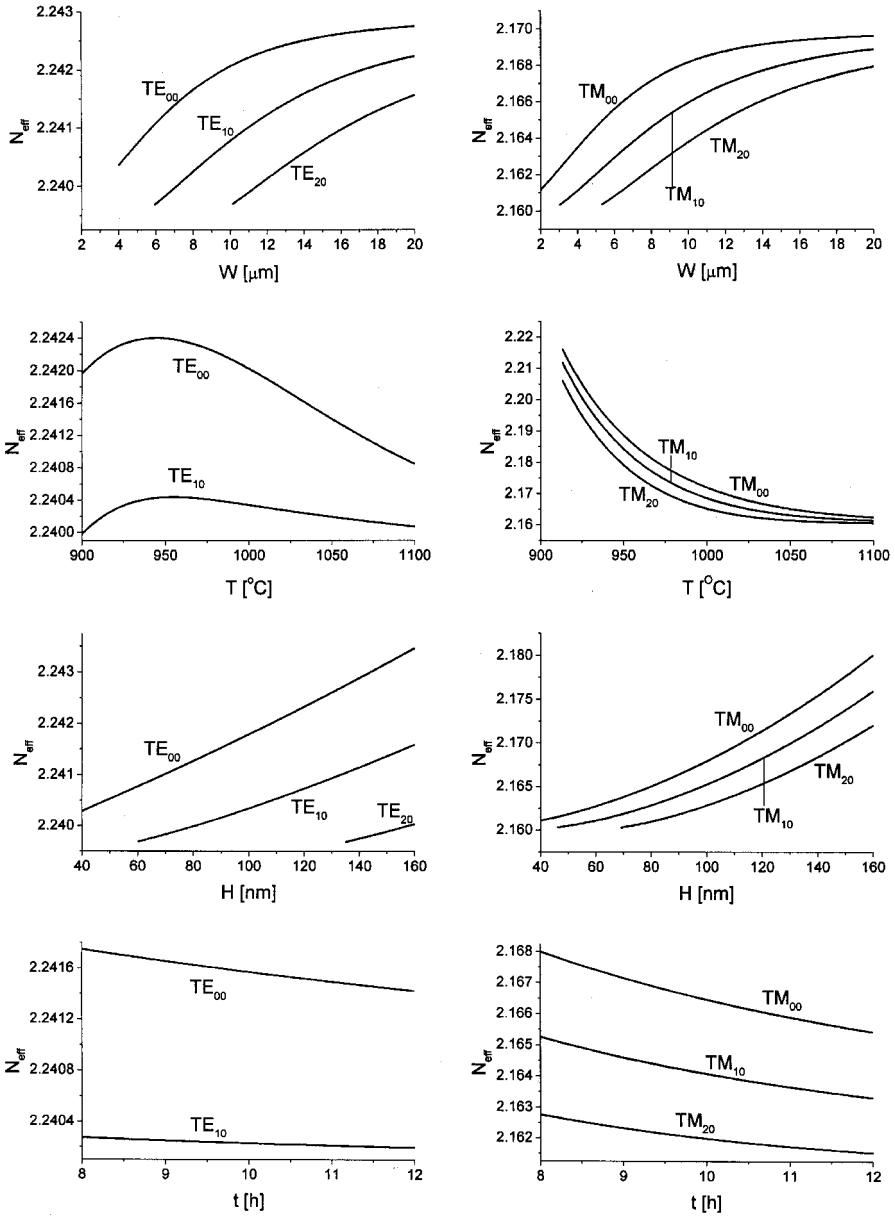


Figure 4. TE and TM mode effective refractive index at 980 nm versus Ti-strip width, diffusion temperature, strip thickness and diffusion time.

pumping in the guide, one must decrease the initial Ti-strip width W or thickness H . In other words, not only the Ti-strip width, as pointed out by Fukuma and Noda [16], but also the Ti-strip thickness and diffusion temperature play an important role in controlling the guided mode number in the waveguide. Figure 5, as an example, illustrates the minimum diffusion temperature which the TE and TM single mode propagation requires, dependent on the wavelength in the range 1450–1650 nm. It can be seen that the minimum diffusion temperature almost

Table 1. TE and TM single-mode conditions at 1532, 1563, 1576, 1480 and 980 nm for z -cut Ti:LiNbO₃ waveguides.

Wavelengths (nm)	modes	W (μm)	T ($^{\circ}\text{C}$)	H (nm)	t (h)
1532	TE	7.0-17.9	≥ 1010	≥ 81	8-12
	TM	3-7.9	≥ 1034	51-94	≥ 9.7
1563	TE	7.4-19	≥ 1020	≥ 88	8-12
	TM	3.1-8.2	≥ 1025	52-97	≥ 8.3
1576	TE	7.7-19.5	≥ 1024	≥ 91	8-12
	TM	3.2-8.3	≥ 1022	53-98	8-12
1480	TE	6.1-16.1	≥ 995	≥ 70	8-12
	TM	2.8-7.4	≥ 1050	48-90	two modes
0980	TE	≤ 6.0	two modes	≤ 60	two modes
	TM	≤ 3.0	three modes	≤ 46	three modes

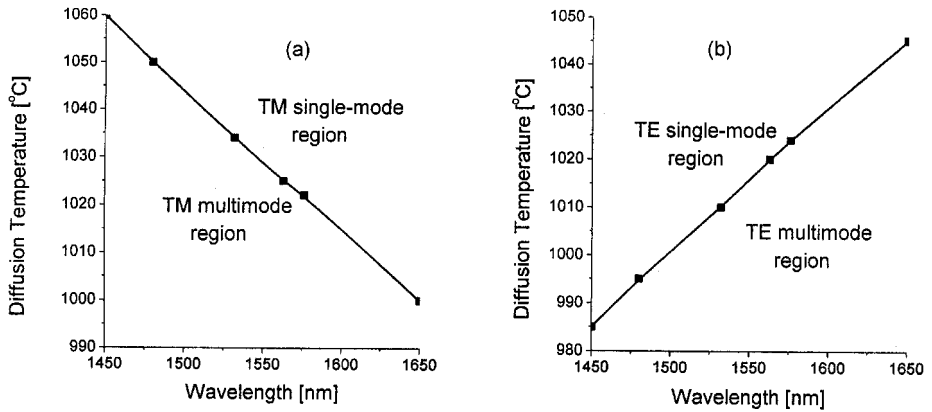


Figure 5. The minimum diffusion temperature that single mode propagation required to be dependent on the wavelength for (a) TM mode and (b) TE mode.

linearly increases and decreases with increasing wavelength for TE and TM polarization, respectively.

2.3. Effective pump area and coupling efficiency

For Er:LiNbO₃ waveguide lasers, the threshold pump power P_{th} and slope efficiency S_e near the threshold can be expressed as [14]:

$$P_{\text{th}} = F \frac{h\nu_p}{\eta_a \sigma \tau_f} \frac{\delta_1}{2} A_{\text{eff}}, \quad (5)$$

$$S_e = F' \frac{T_2 \lambda_p}{\delta_1 \lambda_8} \eta_a \eta_c, \quad (6)$$

where F and F' are the modification factors involving the spatial overlap between the normalized mode intensity distribution and the total Er-dopant concentration profile. δ_1 is the round-trip cavity loss of the fundamental mode. A_{eff} and η_c are effective pump area and coupling efficiency between the laser mode and the pump mode. Under the assumption of Gaussian/Hermite-Gaussian fundamental mode field distribution [14, 15], for the case of coaxial pumping they are expressed as

$$A_{\text{eff}} = \frac{V_{\text{eff}}}{l} = \frac{\pi}{12} \frac{[(W_{xl}^2 + W_{xp}^2)(W_{yl}^2 + W_{yp}^2)]^5}{W_{yl}^2 W_{yp}^2}, \quad (7)$$

$$\eta_c = \frac{3}{5} \frac{W_{xl} W_{yl}^3 [(W_{xl}^2 + 2W_{xp}^2)(W_{yl}^2 + 2W_{yp}^2)]^7}{(W_{xl}^2 + W_{xp}^2)(W_{yl}^2 + W_{yp}^2)^5}, \quad (8)$$

where V_{eff} is the effective mode volume corresponding to the effective pump area, l is the cavity length; W_{il} and W_{ip} ($i = x, y$) correspond to the laser and pump mode sizes, respectively. Hence, using the relation between the mode sizes and the waveguide fabrication parameters established above, A_{eff} and η_c dependency of these parameters can be readily addressed. In figure 6 we show the variations of A_{eff} with these parameters for four possible pumping and output schemes TE-TE, TE-TM, TM-TE and TM-TM, considering the pumping wavelengths 1480 and 980 nm. It is obvious from figure 6 that A_{eff} has a weaker dependence on W or t in their considered ranges, indicating that the Ti-strip width or diffusion time cannot influence significantly the threshold pump power via A_{eff} . On the other hand, A_{eff} increases strongly with increasing diffusion temperature T , and decreases rapidly with increasing strip thickness H , showing that the diffusion temperature and Ti-strip thickness strongly influence the threshold pump power via A_{eff} . For pumping wavelength, either 1480 or 980 nm, there is the relation $A_{\text{eff}}(\text{TM-TM}) < A_{\text{eff}}(\text{TE-TM}) < A_{\text{eff}}(\text{TM-TE}) < A_{\text{eff}}(\text{TE-TE})$, indicating that for a given pump wavelength the TM-TE scheme has the largest effective pump area, while TM-TM has the least value. Moreover, the plots in figure 6 can be divided roughly into four groups according to their values. Group 1 includes the plots of the 980 nm pumped TE-TM and TM-TM schemes and the 1480 nm pumped TM-TM schemes; these plots locate at the bottom of the figure. Group 2 is the 1480 nm pumped TE-TM scheme; group 3 is the 1480 nm pumped TE-TE, TM-TE schemes and 980 nm pumped TE-TE scheme; finally, group 4 is the 980 nm pumped TM-TE scheme, which locates at the top of the figure. Becker *et al.* [2] have reported the following experimental data (pump wavelength 1480 nm): $P_{\text{th}}(1563 \text{ nm, TM-TE}) = 25 \text{ mW}$, $P_{\text{th}}(1563 \text{ nm, TE-TE}) = 22 \text{ mW}$ and $P_{\text{th}}(1576 \text{ nm, TM-TM}) = 13 \text{ mW}$, which yield the ratios $r_1^{\text{exp}} = P_{\text{th}}(1563 \text{ nm, TM-TE})/P_{\text{th}}(1563 \text{ nm, TE-TE}) = 1.14$ and $r_2^{\text{exp}} = P_{\text{th}}(1563 \text{ nm, TM-TE})/P_{\text{th}}(1576 \text{ nm, TM-TM}) = 1.92$, respectively. Under their waveguide fabrication parameters we have the following theoretical data, $A_{\text{eff}}(1563, \text{TM-TE}) = 103.65 \mu\text{m}^2$, $A_{\text{eff}}(1563, \text{TE-TE}) = 95.17 \mu\text{m}^2$ and $A_{\text{eff}}(1576, \text{TM-TM}) = 32 \mu\text{m}^2$. Then we can use equation (5) to evaluate the theoretical values of the above two ratios as: $r_1^{\text{theo}} \approx A_{\text{eff}}(1563, \text{TM-TE})/A_{\text{eff}}(1563, \text{TE-TE}) = 1.09$ and $r_2^{\text{theo}} \approx (A_{\text{eff}}(1563, \text{TM-TE})/A_{\text{eff}}(1576, \text{TM-TM})) (\sigma_{\text{TM}}(1576)/\sigma_{\text{TE}}(1563)) = 2.02$, which are in good agreement with their experimental values of 1.14 and 1.92, respectively.

The calculated results have indicated that, except for the TE-TM scheme, the values for coupling efficiency η_c of the other three schemes vary with these diffusion parameters only over a very narrow range of 0.72-0.85. In figure 7 we only show the variations of η_c with these parameters for the TE-TM scheme for the emission wavelength 1563 nm. It is obvious that in this scheme 1480 nm pumping exhibits lower η_c and higher A_{eff} values (see figure 6), as compared to the 980 nm pumping, and hence yield smaller slope efficiency S_e and larger threshold

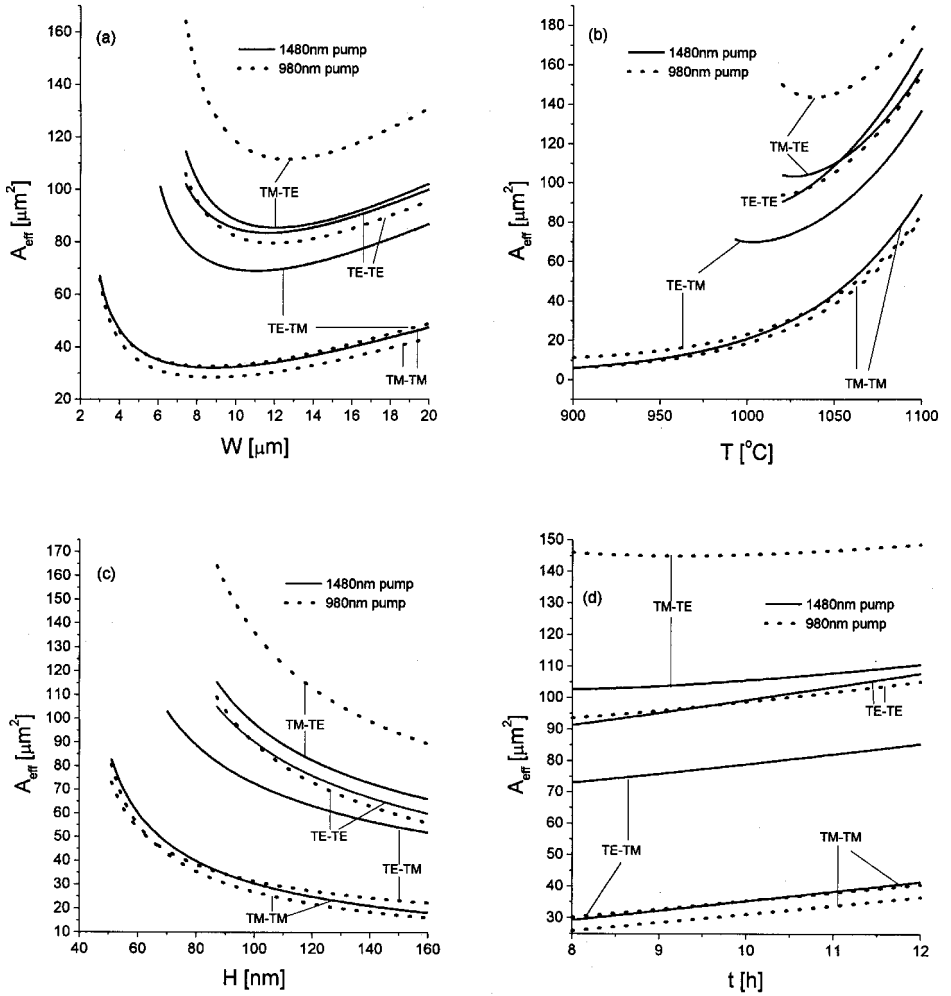


Figure 6. Effective pump area versus (a) Ti-strip width, (b) diffusion temperature, (c) strip thickness and (d) diffusion time under different input-output schemes in the case of lasing wavelength 1563 nm.

pump power P_{th} . In other words, this input-output scheme will exhibit lower laser efficiency. One can see also from figure 7 that diffusion temperature in this scheme has a strong influence on η_c , while the diffusion time t and strip width (especially for 980 nm pumping) do not.

2.4. Coupling loss of a joint between the waveguide and a fibre

The coupling characteristics of a joint between the Er:Ti:LiNbO₃ waveguide and a fibre at the laser output wavelengths 1532, 1563 and 1576 nm and pumping wavelengths 980 and 1480 nm is very useful as it is used as the light source for fibre communication. Their coupling efficiency is characterized by the overlap of guided modes in the fibre and the waveguide [17]. Under the assumption of Gaussian mode field distribution in the fibre and the above used mode profile in the

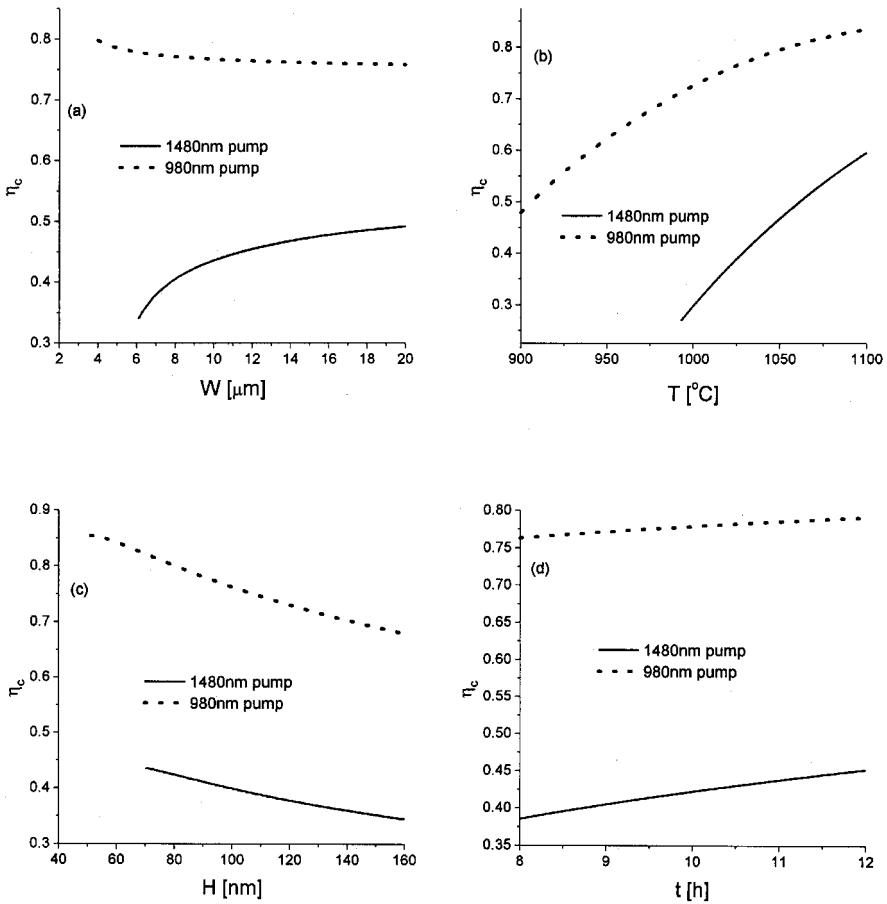


Figure 7. Coupling efficiency of laser and pump modes versus (a) Ti-strip width, (b) diffusion temperature, (c) strip thickness and (d) diffusion time for the TE-TM scheme and the case of lasing wavelength 1563 nm.

waveguide, and of the coincidence of the maxima of the vertical (i.e. y direction) mode field distributions in the fibre and waveguide, the coupling efficiency in the lateral and vertical directions can be expressed as follows:

$$K_x = \frac{2}{\frac{W_x}{W_F} + \frac{W_F}{W_x}}, \quad (9)$$

$$K_y = \frac{8(2^{1/2})W_y W_F^3}{\pi(W_y^2 + 2W_F^2)} \exp\left[-\frac{W_y^2}{2W_F^2}\right] [1 + \alpha\pi^{1/2} \exp(\alpha^2)(1 + \text{erf}(\alpha))]^2, \quad (10)$$

where

$$\alpha = \frac{W_y^2}{2^{1/2}W_F[2(W_y^2 + 2W_F^2)]^{1/2}}, \quad (11)$$

where W_F is the half width at $1/e$ intensity in the fibre and in the following calculation it is approximated as $5/2^{1/2} \mu\text{m}$ for all involved wavelengths [18]. The total coupling efficiency, in the case where the Fresnel reflection loss is ignored, is given by

$$K = K_x K_y. \tag{12}$$

Consequently, the coupling loss L is given by

$$L = -10 \log K = -10 \log(K_x K_y). \tag{13}$$

Figure 8 shows the calculated results of the coupling loss against the diffusion parameters for the two pumping wavelengths and the representative emission

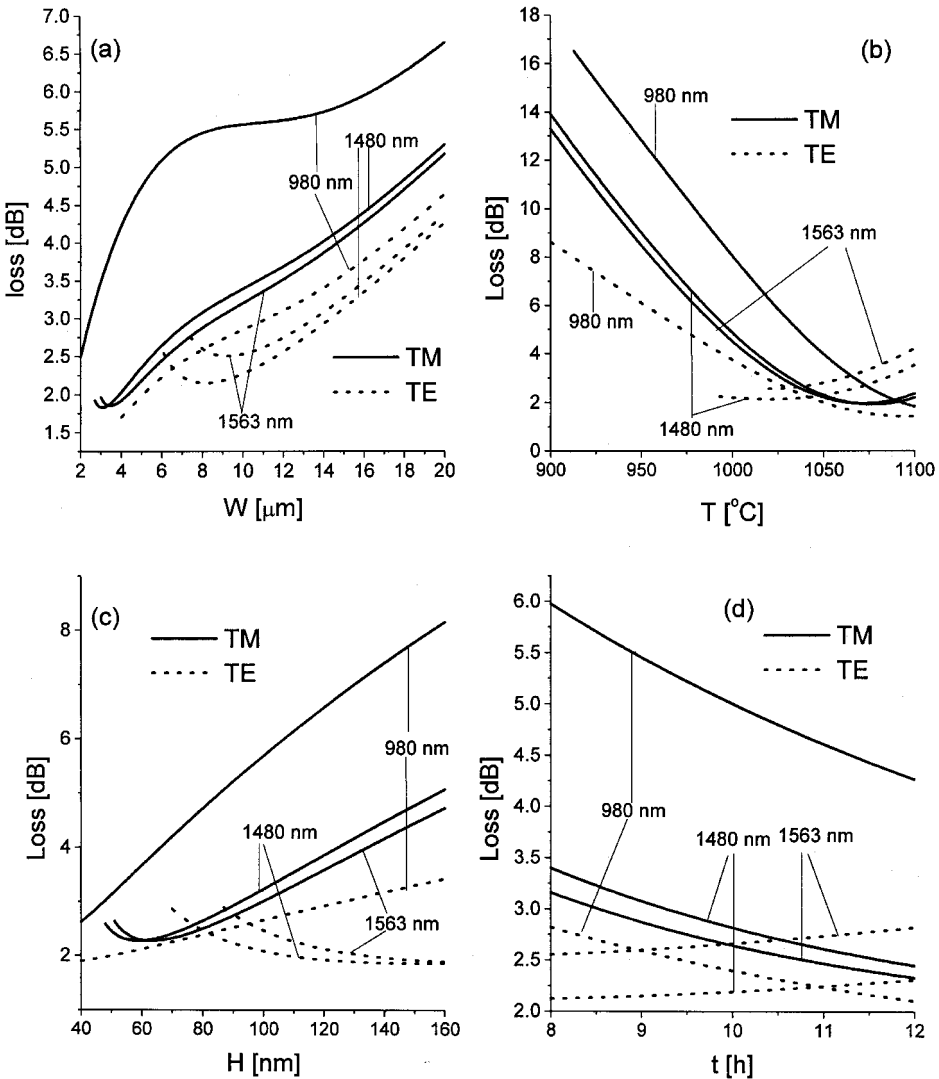


Figure 8. Coupling loss of a joint between the waveguide and a fibre versus (a) Ti-strip width, (b) diffusion temperature, (c) strip thickness and (d) diffusion time for TE and TM modes at 980, 1480 and 1563 nm.

wavelength 1563 nm. For a given wavelength the coupling loss of the TM mode is generally larger than that of the TE mode. Furthermore, for the TM mode, the loss decreases as the wavelength increases; it also exhibits more significant dependence on the diffusion parameters T , H and t than that of the TE mode. When the strip is wide and the Ti film is thick, the coupling loss is large, and the opposite tendency is observed in the plots of loss against diffusion parameters T and t . For TE modes, however, the variation of the loss with these parameters and wavelength is more irregular compared to the TM mode, as shown in figure 8. Fukuma and Noda [16] have reported the experimental results of the coupling loss versus the diffusion temperature at wavelength 1.15 μm , which is close to the 980 nm considered here. Because of the differences in diffusion conditions employed in their experiment and this calculation, it is not possible to give an accurate comparison for them. However, it is possible to make a qualitative comparison. One can see from figure 8(b) that for the wavelength 980 nm, the TM mode coupling losses decrease more rapidly with the diffusion temperature than do the TE mode losses, and this is consistent with Fukuma and Noda's experimental results. In addition, the losses in the temperature range of 975–1050°C vary in the range of 4.89–2.0 dB for the TE mode and 10.4–4.2 dB for the TM mode, and the former is reasonably in agreement with the experimental data, 4.0–2.1 dB. The discrepancy of the latter with the measured value 5.2–1.7 dB can be interpreted by the difference in the diffusion parameters employed. In their experiment the parameters are $W = 10$ or $6 \mu\text{m}$, $t = 5$ h and $H = 50$ nm, while the figures considered here are $8 \mu\text{m}$, 9 h and 95 nm, respectively. The influence of H and t on the loss is not significant for the TE mode, as shown in figures 8(c) and (d), hence the above good agreement between the theoretical and experimental results is observed for the TE mode. On the other hand, the influence of H on the loss is evident for the TM mode and it increases with increasing H , and hence, the above larger prediction of the loss for the TM mode is reasonable.

3. Conclusions

We have demonstrated a detailed comparative analysis of TE and TM modes guided in a *Z*-cut Ti-diffused *Er:LiNbO₃* strip waveguide laser. This analysis considers the possible lasing wavelengths 1532, 1563, 1576 nm and pumping wavelengths 1480 and 980 nm, as well as the possible four pump-output mode polarization schemes. Using the variational technique, mode size and the corresponding effective refractive index, effective pump area, coupling efficiency of laser and pump modes, and coupling loss of a joint between the waveguide and a fibre, all of which are dependent on the controllable waveguide fabrication parameters, were calculated and discussed. The single mode propagation conditions were collected for all of the lasing and pumping wavelengths for both TE and TM modes. In order to examine the accuracy of the analysis, the calculated results were compared with experimental data. The comparisons have revealed good agreement. These theoretical results could offer guidance to the design and fabrication of the device.

Acknowledgments

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